

Making the best possible choice: Using agent-based modelling to inform wildlife management in small communities

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ABSTRACT

Human-wildlife interactions occur where human societies interface with the natural landscape. Some of these interactions can be problematic and lead to negative outcomes, such as the destruction of property or injury to humans or wildlife, especially when they involve large carnivores such as bears. Developing management strategies to mitigate these negative outcomes can be costly, time-consuming, and ethically problematic. We developed an agent-based model of black bears foraging in the forest-urban interface of a small resort community to simulate and test a variety of management strategies. The model allows to quickly and cost-effectively identify those strategies, such as attractant management and deterrence measures, that are most likely to reduce the number of bears that become habituated to human areas and conditioned to anthropogenic food sources. Although increasing the likelihood that a bear is deterred from human areas did have some effect, model outcomes indicated that the elimination of sources of anthropogenic food was the more effective management strategy. However, the model also revealed that interactions between the spatial configurations of these two approaches impacted the overall effectiveness of each. Notably, the random distribution of an increase in deterrence across the area of interest in conjunction with a focus on attractant removal around the perimeter of said area was the most effective spatial implementation of the two strategies together. This result provides a quantitative confirmation that reducing anthropogenic food availability should be the highest priority for managers seeking to reduce the number of conflict bears in urban areas, but also that the spatial configuration of any applied strategies is important to consider. The agent-based modelling approach proved a fast and cost-effective way to compare a variety of possible management strategies and the different ways in which they can be arranged in a specific landscape.

1. Introduction

Human-wildlife interactions are common for species whose habitat requirements overlap with our own. The American black bear (*Ursus americanus*) is one such species; black bears' range extends over much of North America and they favour many habitats that are suitable for human settlements (Garshelis et al., 2016). Understanding how best to reduce negative human-bear interactions for the safety of bears and humans is a key management concern for affected communities.

Black bears are omnivores whose natural diet includes insects, herbaceous vegetation, berries, fish, and mammals. They are also attracted to anthropogenic food sources such as improperly stored garbage, fruit trees, and gardens (Lewis et al., 2015), although black bear foraging on

anthropogenic food sources is more prevalent when high quality natural food is not available (Baruch-Mordo et al., 2014; Obbard et al., 2014). As urban area increases, so does the potential overlap of human and bear territory and therefore the opportunity for a bear to access anthropogenic food sources, such as kitchen scraps or garbage, that have a high caloric value. Bears that obtain such a food reward are at risk of becoming food-conditioned and may learn to associate anthropogenic features with a high-quality food source (Hopkins et al., 2010; Society and McCullough, 1982).

If bears are not deterred or otherwise interdicted from the anthropogenic food source or the anthropogenic features in general, they are at risk of becoming habituated to humans (Appleton, 2006; Elfström et al., 2014; Herrero et al., 2005); they may lose their wariness of urban areas

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and people, making them less likely to flee when they are confronted by a human (Homstol, 2011; Mazur, 2010). Bears that are both habituated to humans and food conditioned (hereafter: conflict bears) are at a high risk of a negative human-bear interaction (Hebblewhite et al., 2003; Herrero and Higgins, 2003; Lewis et al., 2015). Conflict bears may damage human property or crops, and may also pose a threat to pets or humans (Herrero et al., 2011; Herrero and Higgins, 2003; McCarthy and Seavoy, 1994).

Lethal control measures are used in cases where a conflict bear poses a potential threat to the public (Spencer et al., 2007). However, it does not appear that even increased harvest quotas have a measurable effect on the number of human-bear conflicts (Obbard et al., 2014), and the destruction of black bears is not always a socially-acceptable recourse (Baruch-Mordo et al., 2011; Can et al., 2014; Noel and Pienaar, 2017). Further, eliminating conflict bears is likely to impose sink dynamics on the local population (Baruch-Mordo et al., 2014; Beckmann and Lackey, 2008; Hebblewhite et al., 2003). Such a dynamic, where urban areas act as sinks for the general population, could disproportionately affect females in the population. Females with cubs tend to inhabit near-urban areas more frequently than do adult males (Elfström et al., 2014), either due to competition with males or potentially using the proximity to urban areas as a predator shield as has been found in some studies of brown bears (*Ursus arctos*) (Steyaert et al., 2016). This uneven sex distribution may compound the local sink dynamic as cubs of food-conditioned females tend to also forage in urban areas; they are thus more likely to become conflict bears and die due to interactions with people or anthropogenic features (Mazur and Seher, 2008).

Non-lethal responses exist but their long-term effectiveness is questionable. Translocation is sometimes attempted, but continued use of this practice may be due to social pressure rather than to a high rate of success (Spencer et al., 2007). Black bears are intelligent and able to travel long distances; they may return to their home range after a translocation (Appleton, 2006; Elfström et al., 2014). Alternatively, they may be relocated to an already-occupied territory and face higher mortality rates. Another option is aversive conditioning, which attempts to forge in the bear an association between humans and a negative stimulus. This approach can be effective, but it requires persistent application (Mazur, 2010) and cannot reduce the degree to which a bear is food-conditioned already (Homstol, 2011).

In general, strategies like aversive conditioning, translocation, and lethal control are responses to bear behaviour. They inherently address a symptom of the larger problem of human-bear habitat overlap, rather than reducing the potential for negative human-bear interactions. Recent research highlights the need for more proactive strategies like reducing or eliminating food attractants available to bears by means such as public education (Baruch-Mordo et al., 2011; Gore et al., 2006), policy change (Noel and Pienaar, 2017), harvesting or removing urban attractants like fruiting trees and bushes (Lewis et al., 2015), or using active deterrents like electric fencing that excludes bears from strong attractants such as garbage dumps. Such proactive approaches, which can be implemented in anticipation of bear presence, should have some theoretically ideal implementation. However, optimal bear management strategies will differ between locations or communities due to variation in urban and natural landscape configurations (Can et al., 2014); there is no globally optimal implementation strategy for mitigation of negative human-bear interactions. Experimentally determining locally optimal implementations is not possible or sensible due to economic constraints, time and labour demands, and ethical concerns.

Individual- or agent-based modelling allows for the simulation of a variety of different landscapes and management strategies at the level of individual animals (Grimm and Railsback, 2005) and is potentially an effective tool for exploring solutions to mitigating negative human-wildlife interactions. Marley et al. (2017) built a toy model that simulated black bear movement in a fictional urban and natural landscape to evaluate the likelihood of human-bear conflict over a range of

deterrence and food reward configurations and fractions. That framework has been extended to include more realistic bear movement modelling, using the Resort Municipality of Whistler (Whistler), British Columbia, Canada as a case study (Marley et al., 2019). In the present work, we describe an improved and empirically validated version of this agent-based model and use the model to explore the relative effectiveness of bear deterrence, anthropogenic attractant removal, and other management tools in reducing the number of conflict bears in a human community. Our simulations allowed us to compare the overall effectiveness of these different strategies, as well as the interactions and synergies between them, in search of the most effective management approach for Whistler. The results show that managing anthropogenic food sources is the single most important factor in the reduction of human-bear interactions in human communities, but also highlight the need to consider interactions between different spatial configurations of management strategy applications.

2. Study area

The modelled area is in southwestern British Columbia (BC), Canada, roughly 100 km north-northwest of the city of Vancouver, BC. It encompasses the entirety of the Resort Municipality of Whistler (Fig. 1) and represents roughly 188.9 km² of natural and developed landscape.

3. Methods

We used the NetLogo modelling environment (v6.1.1; Wilensky, 1999) to develop a model that simulates the movement and foraging behaviour of an individual bear moving through a landscape that represents the Resort Municipality of Whistler, BC. Empirical data and expert opinion as described below were used to derive landscape characteristics, bear movement rules, and bear habituation rules. The overall model represents the Whistler spatial landscape and can simulate the movement of a single bear based on the real-world bear movement patterns as derived from GPS collar data. Additionally, it is visually easy to understand and communicate the model mechanics and results due to the cartographic representation of the model landscape; see Fig. 2 for a snapshot taken from the working model.

3.1. Model parameter derivation

3.1.1. Landscape characteristics: bear food value

The study area is a mosaic of natural and urban landscapes with varying levels of productivity and food availability. To assign natural food values for the model, landscape areas were ranked using spatial

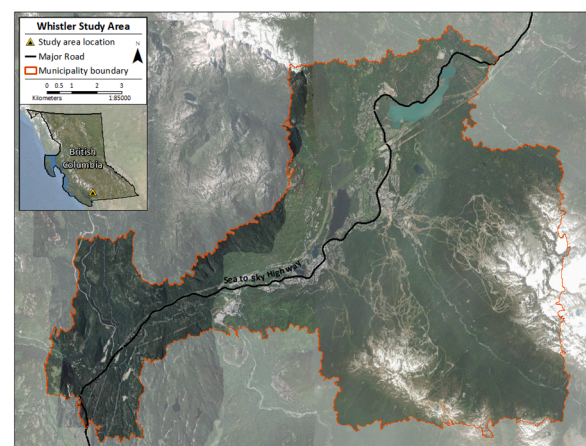


Fig. 1. The study area consisting of the natural and urban landscape contained within the municipal boundaries of the Resort Municipality of Whistler in southwest British Columbia, Canada.

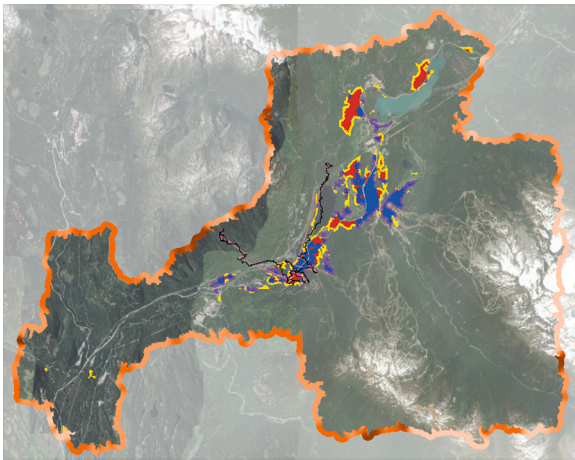


Fig. 2. Example snapshot of the Whistler bear model during a run. Satellite imagery is shown for cells that were assigned a natural food value or zero food value (e.g., glaciers); imagery shown is not a perfect representation of the model landscape as derived from TEM data and is included for visualization purposes only. Orange cells indicate a buffer zone around the municipal boundaries. Red cells indicate urban area with base parameters; blue cells indicate urban areas with increased deterrence; yellow cells indicate urban areas with attractant management; and purple cells indicate urban areas with both attractant management and increased deterrence. The black line traces the path taken by the model bear, and the pink line indicates the straight line between two step ends.

vegetation community composition data from a Terrestrial Ecosystem Mapping (TEM) project (Green, 2004) evaluated with expert consultation from Tony Hamilton, Large Carnivore Specialist for the BC Ministry of Environment, Conservation Science Section. Different groupings of plant species were designated as individual bear habitat types, and final food values for a given bear habitat type depended on the plant species present, elevation, and structural stage as determined by the time since last disturbance. This process, which converts TEM landscape characteristics to a simple relative weighting of the food value as perceived by the bear, would translate to other landscapes in the Province for which TEM data are available. An assessment of another landscape for which TEM data are available using this method would therefore allow the model described in this paper to be applied to that assessed landscape with relatively little additional effort in terms of landscape or data analysis.

A bear's vegetative diet differs seasonally; bears consume more green vegetation early in the late spring and early summer and rely heavily on berries and other mast in the late summer and early fall (Baruch-Mordo, 2012; Coogan and Raubenheimer, 2016). Accordingly, the natural food value was assessed for two independent seasons, designated the pre-berry and berry seasons (Fig. 3).

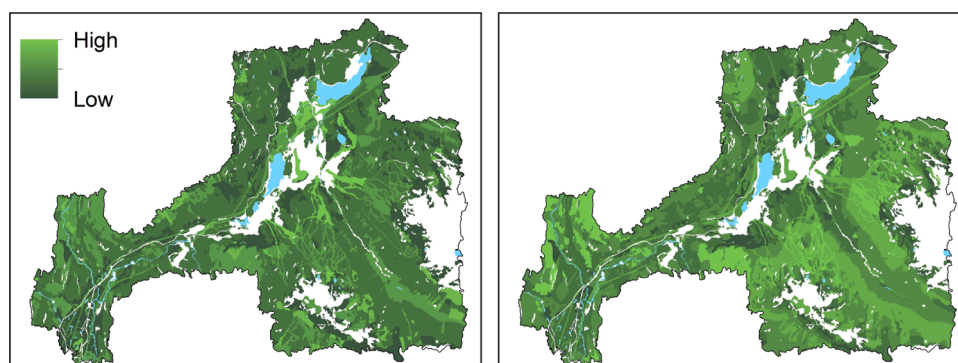


Fig. 3. Food value assessment values for the pre-berry season (left) and berry season (right) in Whistler, BC. White cells have a natural food value of 0.

For urban areas, food value was determined by analysing detailed land use zoning information with a resource selection function (Johnson et al., 2006). Zoning classes visited more often by real bears were assumed to have a higher food value than those visited less frequently. These derived values were scaled to match the natural food values such that maximum and minimum urban food values and natural food values were equal. See Fig. 4 for a model snapshot depicting the initialized model landscape with included natural and urban food values.

3.1.2. Bear movement model derivation

We calculated step length and turning angle distributions from bear movement data from eight bears (seven males, one female) whose movement overlapped the study area (Homstol, 2011) using the statistical language R 3.6.3 (R Core Team, 2020) and the package AdehabitatLT (v0.3.25; Calenge, 2006). Location fixes for the bear movement data used for movement derivation were recorded hourly. The distribution of the step lengths is bimodal (Figure C1-A); the bear step lengths can be divided into short steps of less than 90 m and long steps of greater than or equal to 90 m. We visually assessed the distribution of turn angles for the short and long steps (Figures C1-B, C1-C). Short steps tended to be associated with a 180-degree turn, whereas long steps tended to preserve directionality; turn angles associated with long steps were most frequently between -50 and 50° . These distributions were used to parameterize model bear movement as described below. Including turn angle differentiation in the separation of movement modes allows for an improved approximation of bear behaviour than earlier versions of this model, which associated modelled step lengths with all possible turn angles.

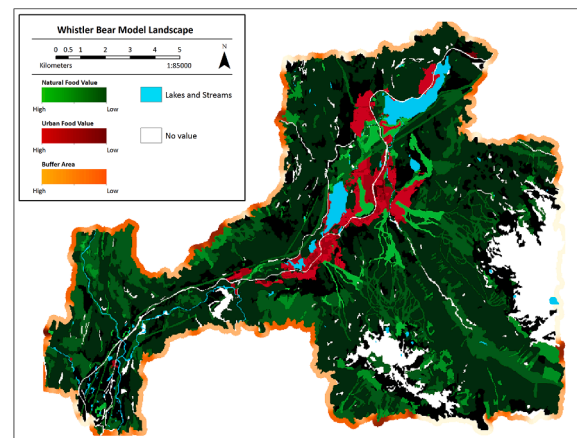


Fig. 4. Example of model landscape after initialization, showing the pre-berry season.

3.1.3. Movement validation

To assess whether the bear movement rules adequately modelled the movement behaviour of real-world bears, we ran a truncated version of the model. For each GPS collar location within the study area boundaries, we used the R package ADEHabitatLT to calculate the heading of the bear at that point based on the bear's angle of travel from its previous location (Calenge, 2015). We loaded these points and their associated heading data into the NetLogo model of the food landscape derived in Section 3.1.1 and used each point and heading combination as the start point from which the model bear could make its movement selection. We then compared the food value of the location chosen by the model bear and the food value of the location chosen by the real bear for each start-choice point pair available in the study area ($n = 6115$).

To ensure that the model bear selection weighting is representative of the selection weighting made by real bears we recorded the food value selected for each start-choice pair in addition to the twenty food values from which that value was chosen by both real bears and model bears. For each such selection array, we scaled all food values to the maximum value in the array, and then ranked the value selected according to its position in the scaled array. With this approach, each value selected by the bear was assigned a ranking proportional to the maximum value available to the bear to obtain a distance value from the food value selected to the maximum food value available. We calculated these distance values for all start-choice pairs and compared the distribution of distance values for real bears to the distribution of distance values for the model bears to evaluate how closely the model bear food value selection mirrored that of the real bears. Validation results for both food value selection and the weighting of that selection are reported in the Movement validation subsection of the Results.

3.2. Model overview

The model processes described in this section serve as an overview in brief. A complete model description following the ODD (Overview, Design concepts, Details) protocol, which is a standard framework for describing agent-based models (Grimm et al., 2020, 2010, 2006), is available in Appendix D.

The purpose of this model is to simulate a black bear moving through the natural and anthropogenic landscape of the Resort Municipality of Whistler. The model compares different management strategies and evaluates their relative effectiveness within the model. Evaluating these strategies and their interactions provides support for actual management decisions. We are specifically comparing the effectiveness of attractant management, increased deterrence, and reactive deterrence in reducing bear presence within urban cells. To ensure that the model is realistic enough to model real bear behaviour in the context of different management regimes, we considered patterns of movement and food selection. We compared real and model bear step length and turn angles, along with each bear's selection of food values in the landscape, to determine that the model bear adequately represents the behaviour of real bears in our landscape.

The behaviour of the model bear is the product of the interaction between movement rules derived from GPS collar data (see Section 3.1.2) and the estimated food value of the landscape (see Section 3.1.1). These movement rules and landscape characteristics inform the following processes that occur at each time step, representing one hour of passed time. First, the bear determines whether it is in an urban or natural area. If it is an urban cell, the bear may be deterred and will leave the area. If the bear is not deterred, it will determine whether it can access food from the current urban area. If it can access urban food, or if it is in a natural area, the bear assesses the food value of the nearby landscape and may either eat or move. If the food nearby is sufficient, the bear will decrease the local food value. If nearby food values are below a threshold, the bear will attempt to move to a higher-value region. See Fig. 5 for a flow diagram that describes the bear behaviour for a single time step.

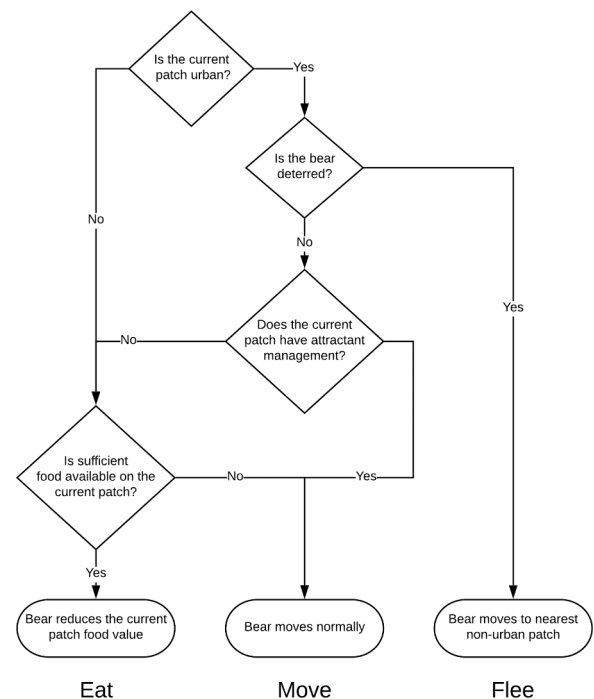


Fig. 5. Flow diagram that describes the decisions made by the model bear at each timestep of the model run. This is a simplified representation of the model processes that determine bear movement; Appendix C2 shows a schema that follows the model processes in detail.

When the bear moves, it generates an array of possible target destinations chosen in accordance with its movement rules; it selects a target destination from this array, with a higher likelihood of choosing a high-value target than one with a low food value. In the original Whistler model (Marley et al., 2019), the bear would move directly to the target destination without interacting with the intervening cells. In this modified version of the model, the bear transits the landscape in an approximation of continuous movement: it moves towards the target cell on a cell-by-cell basis, always moving closer to the target cell but by way of the highest-value intervening cells rather than in a direct straight path (Fig. 6). With this rudimentary pathfinding, the model bear can interact with the landscape on a more continuous scale and can interact with the model landscape as it moves towards its target. This modified movement

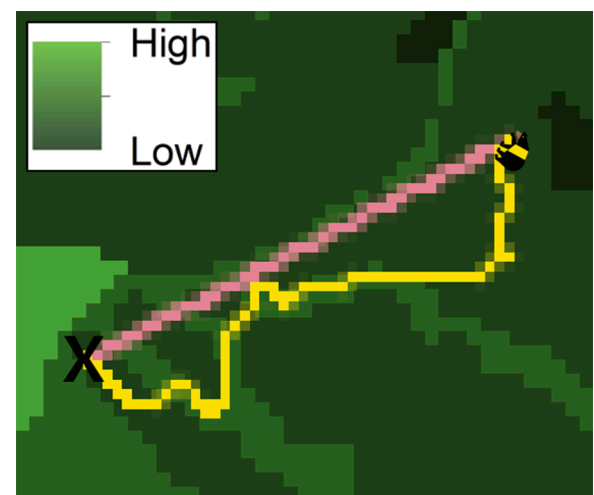


Fig. 6. Example of model bear movement during a single time step. X indicates the starting cell, pink cells indicates the direct path to the target cell, and yellow cells indicate the path taken.

model allows the model to explore aspects of existing management approaches, such as the possibility of mid-movement deterrence, and the exploration of management approaches that could not previously be explored, like reactive deterrence agents or exclusion fencing.

Each time the bear successfully obtains food from an urban area, the bear's food conditioning and habituation levels increase. In contrast, every time the bear is deterred, its habituation decreases. These two values are tracked throughout the model run, and if their combined value exceeds a threshold the bear is considered a conflict bear and the model run terminates; we based this threshold value on a conflict tracking matrix used by conservation officers to determine when a particular bear should be destroyed (BC Ministry of Forests Lands, 2016, Appendix D 2.2, Appendix Figure D1). If the combined value of food conditioning and habituation never reach that threshold level, the model runs for 4800 time steps to simulate 200 days.

The key design concepts of this model pertain to negative human-bear interactions, their causes, and potential ways to decrease these negative interactions. The first design concept is that negative human-bear interactions are due at least partially to bears opportunistically foraging on anthropogenic food sources. Second, there are different management strategies that can have different levels of effectiveness with regard to the reduction of opportunistic foraging and therefore, conflict bears. By modelling bear movement throughout the simulated landscape, it should be possible to determine which management approaches are the most effective.

3.3. Scenario descriptions

3.3.1. Attractant management and deterrence configuration and proportion

We modelled both attractant management and increased deterrence independently to assess the impact of spatial configuration and proportion of the management approach applied. We used three different possible spatial configurations: random, cluster, and perimeter (see Fig. 7 for examples of these configurations). For each of these configurations, we applied attractant management or increased deterrence to a range of proportions of urban cells from 60–100% in 10% increments. For each of these fifteen combinations, we ran the model 100 times for both attractant management and increased deterrence ($n = 3000$).

3.3.2. Attractant management configuration and increased deterrence configuration

To model the interaction between configurations, we ran scenarios with implementations of each spatial configuration of attractant management combined with each spatial configuration of increased deterrence. Attractant management and increased deterrence was applied to 75% of urban cells. Each management approach was implemented independently of the other; the application of one management approach in a cell made the application of the other no more or less likely (see Appendix D: 4.3.2). For each of these nine combinations, we generated 100 replicate landscapes using the same initial parameters

and ran the model once for each replicate ($n = 900$).

3.3.3. Proportion of urban area with attractant management and increased deterrence

We explored the interaction between the proportion of urban cells with attractant management and the proportion of urban cells with increased deterrence. The only configuration for either management approach in this scenario was random (see Appendix D: 4.3.2). For each value from 20–100% in 20% increments, each possible combination of proportion of each management approach was simulated. For each of these twenty-five combinations of starting parameters, we generated 100 replicate landscapes and ran the model once using each of these replicates ($n = 2500$).

3.3.4. Reactive deterrence

To evaluate the effectiveness of reactive deterrence (imitating the role of a conservation officer), we varied the delay and the persistence of reactive deterrence. Increased deterrence and attractant management were each applied independently to 75% of the urban cells and the only configuration for either management approach in this scenario was random (see Appendix D: 4.3.2). Any cells that triggered reactive deterrence increased their deterrence probability by a factor of 2.5. We assessed response delays from zero to three hours and persistence durations from two to four hours and ran 100 simulations of each possible combination of those variables ($n = 1200$).

3.4. Analysis

For all model scenarios described above (see section Scenario descriptions) we fit a key outcome of the model runs, the frequency that the bear becomes a conflict bear, to a binomial logistic model using a generalized linear model (GLM). We used predictor variables as appropriate for the given scenario, and included interaction terms between different management approaches if significant; predictor variables and interaction terms are described in Table 1. We also calculated the proportion of steps taken by bears that ended in an urban cell for all scenarios evaluated and modelled that proportion as a function of the predictor variables relevant for a given scenario; these values were transformed

4. Results

4.1. Movement validation

There is reasonable agreement between cell selection in the food landscape by real and simulated bears for both berry and pre-berry seasons; although there are differences between real and simulated bears, the densities of selected food values have similar peak values and the same overall distribution shape, suggesting that the underlying patterns of food selection are the same (Fig. 8A).

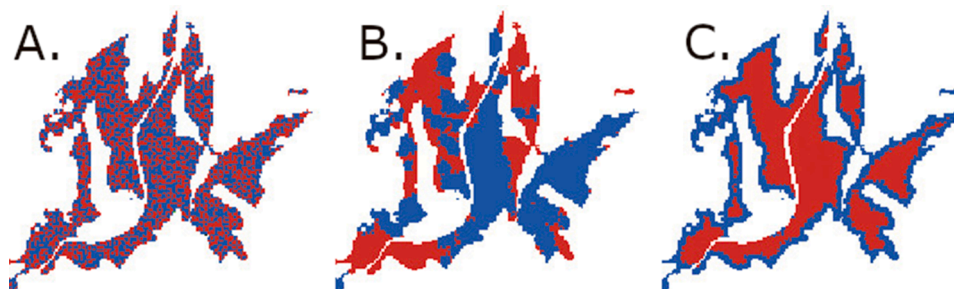


Fig. 7. Example configuration applications of management strategies. In this example image, areas in red indicate an area without management and blue areas indicate an area where management has been applied. Three configuration applications are shown: random application (A), cluster application (B), and perimeter application (C).

Table 1

Overview of terms included in the different generalized linear models used to analyse outcomes from different simulation scenarios. The same predictor variables were used to model both the proportion of bears that became conflict bears in a given scenario (CBM) and the number of steps that ended in an urban cell (ECM).

Scenario name	Predictor Variables	Interaction terms included in CBM	Interaction terms included in ECM
Attractant management configuration and proportion	Proportion of attractant management applied; configuration of attractant management	Yes	No
Increased deterrence configuration and proportion	Proportion of increased deterrence applied; configuration of increased deterrence	No	No
Attractant management and increased deterrence configuration	Configuration of attractant management; configuration of increased deterrence	Yes	Yes
Proportion of urban area with attractant management and increased deterrence	Proportion of attractant management applied; proportion of increased deterrence applied	Yes	No
Reactive deterrence	Response delay; response persistence	No	No

Model bears and real bears appear to have similar non-optimal foraging behaviour when selecting food values from those available. Model bears selected the highest value food cell available roughly 42% of the time; collared bears were only marginally more likely to select the highest value food cell at roughly 46% of the time. The rank distribution of the remaining values selected by both real and model bears is similar (Fig. 8B).

4.2. Attractant management configuration and proportion

The proportion of urban area with attractant management and the spatial configuration of the attractant management applied both affected the likelihood that a bear would become a conflict bear over the

course of a model run (Fig. 9A). The proportion of urban area with attractant management has a significant negative effect on the probability of a bear becoming a conflict bear (Table 2). The perimeter configuration appears to be less effective than either the random or cluster configurations; it results in a significant increase in the proportion of conflict bears generated. However, the interaction between the perimeter configuration and the proportion of managed area lessens the magnitude of this increase.

The proportion and configuration of attractant management also affected the number of steps that ended in an urban cell (Fig. 9C). While the proportion of managed cells was still highly significant (Appendix E1), the difference between management configurations was lower, although the perimeter configuration was still significantly more likely to produce conflict bears.

4.3. Increased deterrence configuration and proportion

Increased deterrence proportion and configuration had a less dramatic effect on the model bear's final status compared to different values of attractant management (Fig. 9B). The proportion of urban cells with increased deterrence had a significant effect on the proportion of bears that became conflict bears (Table 3). However, neither the cluster nor the perimeter configurations impacted the final status of the conflict bear differently from the random configuration.

The proportion and configuration of increased deterrence had very little effect on the fraction of bear steps that ended in an urban cell (Fig. 9D). The proportion of managed cells was still significant (Appendix E2), but the magnitude of the effect was small. There was no significant difference between the random and cluster or random and perimeter configurations of increased deterrence.

4.4. Attractant management configuration and increased deterrence configuration

The interactions between the configurations of attractant management and increased deterrence were unpredictable; some of the combinations were far more effective at reducing the likelihood of a conflict bear than others (Fig. 10A). The most effective combinations used a random configuration of increased deterrence with either a cluster or perimeter configuration of attractant management.

Both the cluster and perimeter configurations of attractant management significantly lowered the proportion of conflict bears, but the configuration of increased deterrence had no apparent effect (Table 4). However, some configurations of increased deterrence did have a

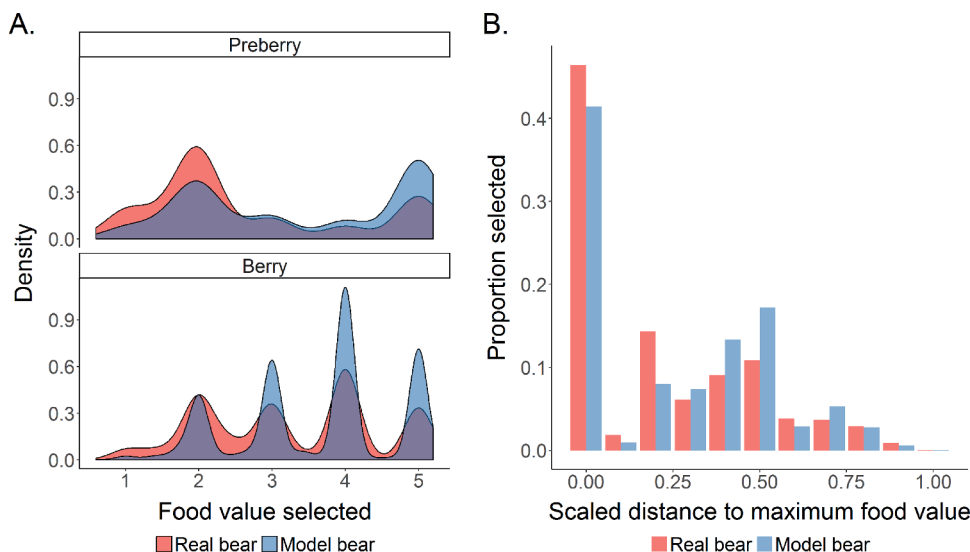


Fig. 8. Validation of model bear movement and food value assessment. A) Density plot showing the food values selected by real and model bears during the preberry and berry seasons. Overlapping area is indicated by a mix of the colours for real and model bears. B) Barplot showing the proportion that each rank interval of food values was selected by both real and model bears ($n = 6115$). The value of the interval indicates the scaled distance from the selected food value to the maximum food value available for that selection; an interval of 0 indicates that the maximum available value was chosen by the bear.

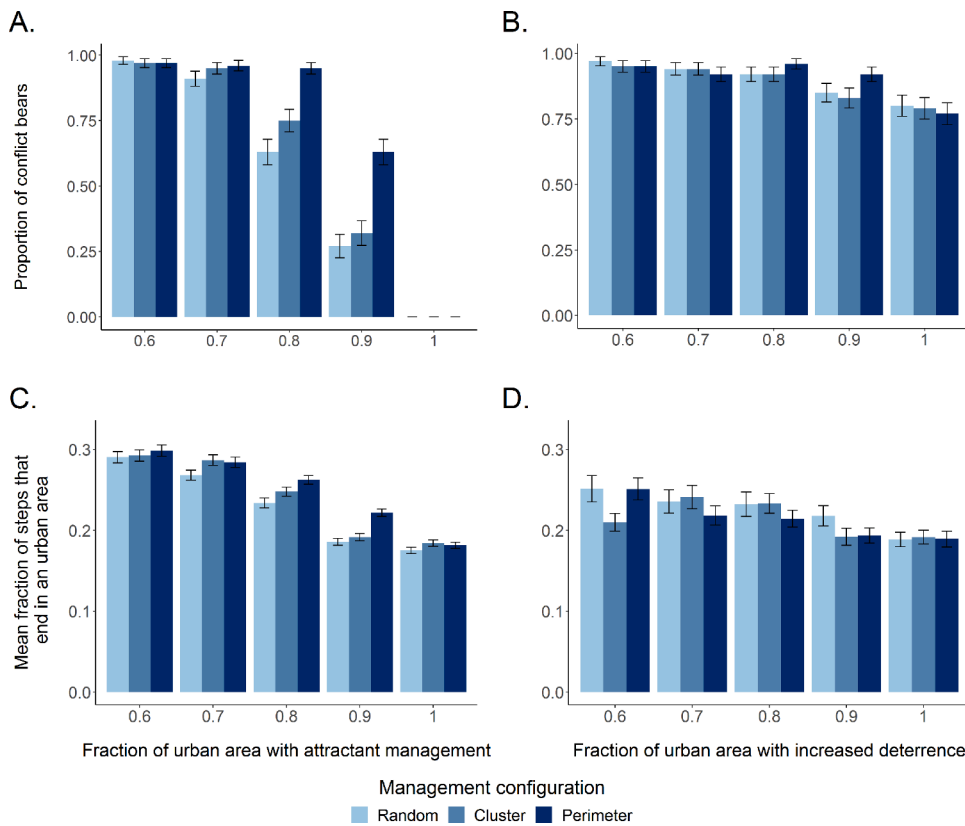


Fig. 9. Barplots displaying the effects of attractant management and increased deterrence on the proportion of conflict bears produced or on the number of steps ending in an urban cell. A) The mean proportion of conflict bears for each combination of urban proportion with attractant management and the configuration of that proportion. B) The mean proportion of conflict bears generated for each combination of urban proportion with increased deterrence and the configuration of that proportion. C) The fraction of bear steps that end in an urban cell for each combination of urban proportion with attractant management and the configuration of that proportion. D) The fraction of bear steps that end in an urban cell for each combination of urban proportion with increased deterrence and the configuration of that proportion. Error bars indicate standard error of the mean ($n = 100$ for each combination).

Table 2

Results of GLM with Cauchy-linked binomial logistic model for bear status predicted by proportion of urban area with attractant management and the configuration of that attractant management. GLM fit: Fisher scoring iterations = 8; residual deviance = 930.21 on 1494 degrees of freedom; AIC = 942.21.

Coefficient	Estimate	β	Std. Error	t-value	p-value
Intercept	19.436	0.000	2.841	6.842	< 0.001
Management proportion	-23.505	-6.846	3.486	-6.743	< 0.001
Cluster configuration	3.037	2.949	4.315	0.704	0.482
Perimeter configuration	50.515	49.043	20.114	2.511	0.012
Proportion: Cluster	-2.586	-2.055	5.102	-0.507	0.612
Proportion: Perimeter	-53.770	-42.730	22.397	-2.401	0.016

Table 3

Results of GLM with Cauchy-linked binomial logistic model for bear status predicted by proportion of urban area with increased deterrence and the configuration of that attractant management. GLM fit: Fisher scoring iterations = 7; residual deviance = 949.01 on 1496 degrees of freedom; AIC = 957.01.

Coefficient	Estimate	β	Std. Error	t-value	p-value
Intercept	14.524	0.000	2.213	6.562	< 0.001
Increased deterrence proportion	-13.267	-6.129	2.269	-5.848	< 0.001
Deterrence: Cluster	-0.146	-0.225	0.394	-0.370	0.711
Deterrence: Perimeter	0.089	0.137	0.427	0.208	0.836

significant interaction with certain configurations of attractant management. For example, the interaction between clustered attractant management and clustered increased deterrence had an overall

significant effect that reduced the effectiveness of either management configuration independently (Fig. 10A, Table 4).

The configuration of management strategies had little impact on the proportion of steps that ended in an urban cell. The most effective combination of strategies was a random implementation of increased deterrence coupled with a perimeter configuration of attractant management. However, the difference in the proportion of conflict bears generated between different strategy combinations was less than 5%. In fact, none of the configurations for either management strategy appeared to have a significant effect (Appendix C3A, Appendix E3), and only the interaction between perimeter configurations had any degree of significance at all.

4.5. Proportion of urban area with attractant management and increased deterrence

Unsurprisingly, the proportion of urban area with attractant management and the proportion of urban area with increased deterrence had an additive effect (Fig. 10B); the higher the proportion of each management approach, the lower the likelihood of a bear becoming a conflict bear. However, the proportion of urban area with attractant management had a much stronger effect that the proportion of urban area with increased deterrence. The mean proportion of conflict bears generated did not fall below 60% without having both increased deterrence and attractant management applied to at least 60% of the urban area. The proportion of each management strategy applied had a significant effect overall (Table 4). The interaction between the proportions was significant as well; as either increased, the overall effect of the other decreased.

The effect of the proportions of increased deterrence and attractant management on the proportion of steps that end in urban cells is less predictable. There is some decrease in that proportion of steps as the urban area with increased deterrence or attractant management increases (Appendix C3B), but there is no obvious trend. Overall, the

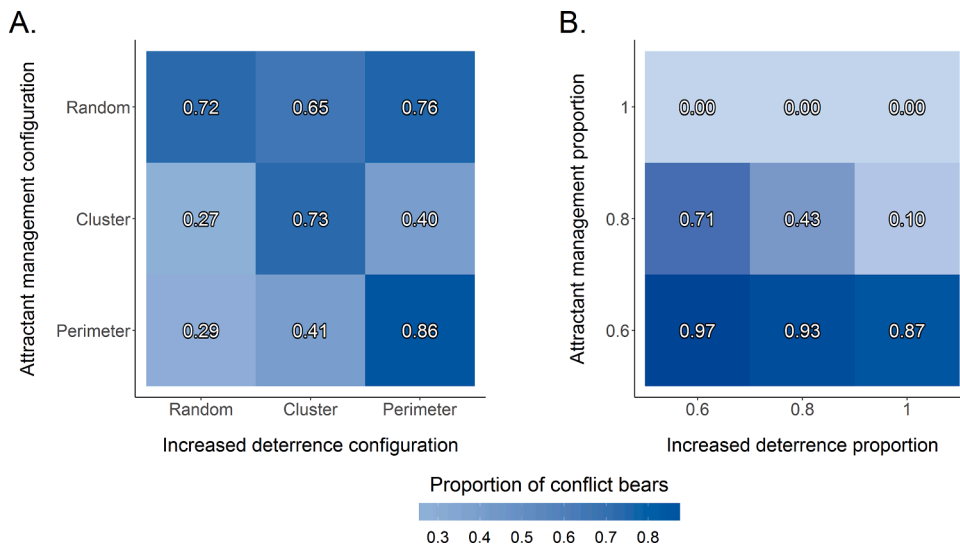


Fig. 10. Tileplots exploring the effect of interaction between attractant management and increased deterrence on the proportion of conflict bears produced. A) The mean proportion of conflict bears generated for each combination of attractant management and increased deterrence configuration. Attractant management and increased deterrence were each applied to 75% of the urban cells irrespective of the application of the other strategy ($n = 100$ for each combination). B) The mean proportion of conflict bears generated for each combination of attractant management and increased deterrence proportion. The random spatial configuration was the only configuration for either strategy ($n = 100$ for each combination).

Table 4

Results of GLM with Cauchy-linked binomial logistic model for bear status predicted by the configuration of attractant management and the configuration of increased deterrence. GLM fit: Fisher scoring iterations = 5; residual deviance = 1063 on 891 degrees of freedom; AIC = 1081.

Coefficient	Estimate	β	Std. Error	t-value	p-value
Intercept	0.827	0.000	0.239	3.464	0.001
Attractant management cluster	-1.709	-1.625	0.345	-4.952	< 0.001
Attractant management perimeter	-1.603	-1.524	0.331	-4.840	< 0.001
Increased deterrence cluster	-0.318	-0.302	0.305	-1.042	0.298
Increased deterrence perimeter	0.238	0.226	0.374	0.635	0.525
Attractant management cluster: Increased deterrence cluster	2.081	1.319	0.466	4.466	< 0.001
Attractant management perimeter: Increased deterrence perimeter	0.803	0.509	0.417	1.925	0.055
Attractant management cluster: Increased deterrence perimeter	0.319	0.202	0.481	0.664	0.507
Attractant management perimeter: Increased deterrence perimeter	2.663	1.688	0.747	3.566	< 0.001

proportion of increased deterrence appears to have a stronger effect, but the maximum difference between combinations is minor (~12%). Although the magnitude of the effect for either proportion was minor, for both attractant management and increased deterrence, the effect of the proportion applied had a significant effect on the number of bear steps that ended in an urban cell (Appendix E4).

4.6. Reactive deterrence

A delayed increase in deterrence probability had a marked impact on the likelihood of a conflict bear (Fig. 11); additional delay time between bear arrival and the reactive deterrence linearly increased the proportion of conflict bears generated. The persistence of the reactive deterrence had little to no effect within groups. This visualization is reinforced in Table 5: the delay period has a large, significant effect whereas the persistence of reactive deterrence has a minor and non-significant effect (Table 6).

The fraction of steps that ended in an urban cell were similarly

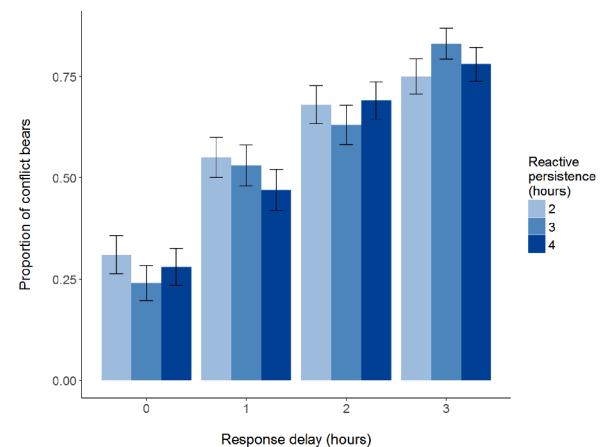


Fig. 11. Barplot of the mean proportion of conflict bears for each combination of reactive deterrence delay and reactive deterrence persistence ($n = 100$ for each combination). Error bars indicate standard error of the mean.

Table 5

Results of a GLM with Cauchy-linked binomial logistic model for bear status predicted by proportion of urban area with increased deterrence and the configuration of that attractant management. GLM fit: Fisher scoring iterations = 9; residual deviance = 974.4 on 2496 degrees of freedom; AIC = 982.4.

Coefficient	Estimate	β	Std. Error	t-value	p-value
Intercept	69.746	0.000	17.002	4.102	< 0.001
Attractant management fraction	-81.114	-49.970	21.232	-3.820	< 0.001
Increased deterrence fraction	-51.009	-31.424	18.350	-2.780	0.005
Attractant management: increased deterrence	55.368	30.508	22.960	2.411	0.016

affected, although to a lesser degree (Appendix C4). The response delay did impact the fraction of steps, but only by around 5% over a range of zero to three hours of delay; although this effect was of a low magnitude, it was nevertheless significant (Appendix E5). The persistence duration had no perceptible significant effect.

Table 6

Results of a GLM with Cauchy-linked binomial logistic model for bear status predicted by the delay and persistence of a reactive deterrence increase in response to bear presence. GLM fit: Fisher scoring iterations = 5; residual deviance = 1464.08 on 1197 degrees of freedom; AIC = 1470.08.

Coefficient	Estimate	β	Std. Error	t-value	p-value
Intercept	-0.568	0.000	0.240	-2.367	0.018
Delay until reactive deterrence	0.683	1.539	0.068	10.084	0.000
Persistence of reactive deterrence	-0.054	-0.089	0.074	-0.734	0.463

5. Discussion

5.1. Bear movement and food landscape validation

Modelling the behaviour and interactions of intelligent animals that learn and display individual behaviour patterns is challenging. If the behaviour model is oversimplified, model output will neither reflect the actions of such animals in the real world, nor will it be able to provide meaningful implications for management. Conversely, a movement model can be over-fitted for a single individual, in which case it would not be applicable to other local animals and could not be used to generate useful management recommendations. We used the optimality of foraging behaviour as a metric for comparison between real and model bears; the close approximation of average model bear behaviour to average real bear behaviour with regards to the optimal selection of food values on the landscape provided confirmation that the model was performing adequately even with simple movement and food-selection rules (see Section 4.1).

Although the movement of model bears was a close approximation of the movement of the real bears used to derive the movement model (see Section 3.1.2), the number of real bears for which GPS collar data were available in the study area was limited. With only eight total bears, and an unequal distribution of sexes (seven males, one female), it is unlikely that the movement model so derived is representative of bears generally, and we are unable to confidently say that the movement model used in this agent-based model would adequately predict the movement of female bears within the study area.

5.2. Management strategy exploration

5.2.1. Attractant management

The results of simulations exploring the relationship between conflict bears and the proportion of urban area with attractant management confirmed the effectiveness of attractant management as a strategy for reducing the number of conflict bears in a human community. The more urban area with attractant management implemented, the fewer conflict bears were generated. The removal of attractants serves to functionally reduce the overlap between human and bear habitat. Decreasing the intersectional area of these habitats where interactions are likely should lead to a reduction in interactions between species and, accordingly, so too should the number of problematic interactions that lead to conflict bears. In reality, increasing the amount of attractant management could be achieved by any number of strategies: increasing the number of bear-proof bins publicly available, strictly enforcing management bylaws, using fencing to exclude bears from gardens and waste sites, restricting or removing native fruit-bearing plants from urban parks and gardens, and a variety of other methods. The overall message from these simulations is a confirmation of a common literature consensus: lower availability of high-calorie and highly-attractive anthropogenic food leads to fewer food-conditioned bears (Baruch-Mordo, 2012; McCarthy and Seavoy, 1994; Peine, 2001). Model results also mirror real world systems where attractant management has been improved and negative human-bear interactions have subsequently decreased (Madison, 2008;

Schirokauer and Boyd, 1995).

The spatial configuration of attractant management also played a significant role, especially at higher levels of attractant management application (Fig. 7A). The perimeter configuration was notably less effective than either the random or cluster configurations, especially when the attractant management strategy was applied to nearly the entire urban area. This may be due to model bears that pierce the management perimeter having free access to anthropogenic food located in the interior urban areas; short of a deterrence event, such bears would have little reason to seek food from non-urban sources and would quickly develop food conditioning. Like model bears, real bears could easily bypass a perimeter of attractant-managed cells either by traversing them directly or by using riparian or other green corridors; the perimeter configuration is likely not the optimal spatial configuration of this management strategy.

The amount or configuration of attractant management had a less obvious effect on the proportion of bear steps that ended in an urban cell than on the proportion of conflict bears overall. However, there are some interesting differences in configuration effectiveness at different application proportions of attractant management (Fig. 7C). For example, when 70% of the urban cells had attractant management applied, the random configuration was less effective in reducing the step proportion than either the perimeter or cluster configurations. At any higher proportion, the random and cluster configurations were both more effective than the perimeter configuration. This inflection may imply that at lower rates of attractant management, the perimeter configuration does have some limited success in terms of deflecting foraging bears at the natural-urban interface. However, once the management is applied to a high enough proportion of urban area, the bears that penetrate the perimeter configuration can wander freely in the urban interior at a rate that is not possible when either the random or cluster configuration is applied. This reinforces that the perimeter configuration is not an ideal implementation for attractant management.

5.2.2. Increased deterrence

Increased deterrence in the model was much less effective than attractant management. Increasing the urban area with increased deterrence did significantly reduce the proportion of conflict bears generated (Fig. 6B), but the magnitude of reduction was limited to around 20% fewer conflict bears when the increased deterrence strategy was applied to the entire urban area. Increased deterrence had a similar effect on the proportion of steps that ended in an urban cell. Although the effect was again significant, the percentage change from 60% to 100% application of increased deterrence was about 5% (Fig. 7D). The draw of high-value anthropogenic food and the associated food-conditioning was enough to overcome any avoidance behaviour due to the increased likelihood of deterrence.

This outcome of relatively low impact of increased deterrence parallels results from studies that found little long term success in deterring bears from accessible anthropogenic food sources, even with the use of active aversive conditioning measures intended to cause an association between humans and pain or noise stimuli for treated bears (Homstol, 2011; Leigh and Chamberlain, 2008). It is possible that frequent and repeated applications of such techniques may be successful with bears that have not already been exposed to anthropogenic food (Mazur, 2010); an improved version of this model could accommodate differences in deterrence effectiveness between naïve and human-aware bears, along with a differentiation between deterrence measures such as low-impact noisemakers compared to pain-causing stimuli such as rubber bullets.

5.2.3. Interaction between the two main management strategies

The interaction between the application proportion of attractant management and increased deterrence was not surprising: the higher the percentage application of both strategies, the fewer conflict bears were

produced (Fig. 8B). Attractant management is clearly more impactful, but the trend direction in the proportion of conflict bears generated is the same for either strategy. Interestingly, there was no large interaction effect between the amount of each strategy applied on the proportion of steps ending in urban cells (Appendix C2B); model bears could always attempt to enter urban areas, regardless of the deterrence measures in place or the accessibility of food. This is likely true of real bears as well, as they are capable of smelling food from great distances and can travel quickly; without a complete barrier around urban areas, bear forays in search of anthropogenic food will likely decrease to a certain point and then plateau, rather than stopping entirely.

The interactions between the spatial configurations of the strategies are less clear. There is no combination of like configurations that is the most effective of the options for a given configuration, for example cluster attractant management and cluster increased deterrence; one of the heterogeneous combinations is always more effective than a like pairing (Fig. 8A). Only in the case of randomly-applied attractant management and perimeter-applied increased deterrence is the combination of matching configurations not the least effective. This trend is most clear when each strategy is applied in the perimeter spatial configuration; this combination of strategies generated the highest proportion of conflict bears. Since the perimeter strategy was in general less effective for both attractant management and increased deterrence when tested on their own, it follows that any bears that penetrate the managed perimeter would be free from either attractant management or increased deterrence measures.

The perimeter configuration performed poorly for each strategy independently and in combination, it is surprising that one of the combinations of strategies most effective at reducing the proportion of conflict bears was perimeter-applied attractant management in combination with randomly-applied increased deterrence (Fig. 8A). This outcome is likely due to synergy between the configurations. First, foraging model bears are less likely to be attracted to the attractant-managed border since food is not accessible on those cells, compared to food readily available on nearby natural cells. Second, any bears that do manage to penetrate the attractant-managed border of urban areas have a high likelihood of being deterred even within the urban core since increased deterrence is applied randomly throughout. A similar synergy is probably occurring in the case of the other most-effective configuration combinations of clustered attractant management and random increased deterrence. Large clusters of urban area, including segments of the border, are unappealing to model bears due to attractant management; any bears that move into a cluster without attractant management are likely to be deterred to the nearest natural cell due to the randomly configured increased deterrence.

5.3. Reactive deterrence

We used response delay and persistence of that response as a proxy for real-world reactive agents that respond to bear sightings and actively seek out the bear to deter it, such as conservation officers, park rangers, or trained dog teams. It appears that the key factor in the effectiveness of such a response is its rapidity; for both the proportion of conflict bears generated and the proportion of steps that ended in an urban cell, the persistence of the response had no bearing on the outcome (Table 5). A rapid reactive response was highly effective in reducing the proportion of conflict bears generated, but the decay in effectiveness was swift. For each additional hour of delay between a bear entering an urban area and the local increase in deterrence likelihood, the proportion of conflict bears increased by 15–25%. Quick response times in the model are analogous to having sufficient personnel available to respond to bear sightings. To effectively employ reactive deterrence in a real-world scenario, the delay between a bear sighting and the arrival of deterrence agents should be made as short as possible through the employment of enough trained staff to respond quickly to bear sightings. Alternatively, this strategy would likely work well as a support to

passive strategies like attractant management.

5.4. Comparison to previous model

The major modifications in relation to the previously published model (Marley et al., 2019) pertain to movement: the movement model now incorporates multiple modes of movement (foraging and travel) for both turn angles and step lengths for the model bear; the model bear now travels across the landscape in a functionally continuous way rather than ignoring potential barrier features like lakes, glaciers, etc.; and the model bear moves in an indirect path to simulate natural pathfinding through a landscape varied in perceived food value. Even with these modifications, some of the key results are similar across both models. For example, both models indicate that preventative action such as limiting bear access to anthropogenic food sources is the most effective approach to reduce the number of conflict bears generated and that perimeter configurations of management strategies tend to be less effective than either random or clustered configuration. This may indicate that in cases where time or resources are limited, using a simpler movement model may give adequate or equivalent results as a more complex movement model. In this paper we extended the prior analyses and incorporated examination of the interaction between spatial configurations and explored the additional management approach of reactive deterrence that would not have been suitably explored without continuous movement.

5.5. Future strategy exploration and model improvement

Bear movement in the model presented here is simplified; the bears follow basic movement rules as they seek out potential target cells with sufficient food. Accordingly, the evaluation of food values and the implementation of those values into the model landscape is critical. The food values in this model are likely conservative; they are based substantively on the assessment of vegetation inventories (Green, 2004) and rely on expert opinion. Ideally, food values in future models could be based on step-selection functions (Thurfjell et al., 2014) or other similar approaches and evaluate multiple landscape characteristics to better approximate the landscape as perceived by bears. Given sufficient collar location data, such an approach could be used to determine urban and natural food values concurrently. Since urban food values are scaled relative to natural food values in this model, it is possible that model bears are undervaluing urban food which is typically of a high-caloric value.

Model bears do not learn in a meaningful or impactful way, and since memory is implemented on a cell-by-cell basis, the likelihood that a model bear's prior actions will affect its later decisions is low. Additionally, for some potential management strategies, limited data are available that describe bear response to those strategies. For example, high-resolution GPS collar data or other tracking records for bear interactions with electric fencing are absent or limited in the literature. For these reasons, more complicated management strategies than those described in the results section, or strategies that rely on more advanced artificial bear intelligence, were not successfully implemented in this iteration of the Whistler bear model.

A variety of large-scale fencing strategies were attempted, with mixed results; see Appendix D 4.3.7 for an implementation based on available movement data. As expected, implementing a complete barrier around any urban areas was 100% effective in reducing the probability of conflict bears, as this approach completely excluded model bears from any anthropogenic food. We tested more practical and nuanced approaches as well, such as strategically placing linear or curved barrier fences across key movement corridors as determined from GPS paths taken by real bears, or placing fences at intervals around the entirety of the urban area. None of these more specifically formulated fences reduced the likelihood of the generation of a conflict bear; in some cases a specific fencing implementation actually increased the proportion of

conflict bears generated. In such cases, the bears appeared to pass through a gap in the fencing and then would often bounce back if they attempted to leave the urban area; these bears were essentially trapped in or near urban areas by the fencing intended to keep them out. It seems likely that real bears would be able to navigate such obstacles more successfully, whether by scent or memory; thus, conclusions from the fencing implementations were not expected to translate well to real bears.

We also explored management strategies that employed vegetation modification to alter bear behaviour as a proxy for techniques like supplemental feeding (Partridge et al., 2001; Robbins et al., 2004). We tried a variety of modifications to the food landscape, examples of which include: the reduction of natural food values in a band encircling any urban areas; the increase of natural food values in a band placed far from urban areas; and point increases of the natural food value. However, due to the limited per-step potential steps, there was little likelihood that a model bear would be affected by these modifications. Only in extreme and unrealistic cases, such as a 200 m wide band of enriched natural food, was the model bear effectively diverted from urban areas.

These unsuccessful attempts to investigate an alternative management strategy highlight the need for behavioural models that accurately reflect the actions of intelligent animals that can learn and remember rather than simply following basic movement rules. Improved sensing and diffuse memory are two components that could be incorporated into this model to explore more complicated management strategies.

6. Conclusion

The principal outcome of this study is a quantitative confirmation that the reduction of anthropogenic food available as forage is the single most important factor to consider in the management of black bears. Other approaches, such as increasing the likelihood that a bear will be deterred from a site with human food, do influence the probability of the generation of a conflict bear, but the magnitude of this influence is relatively small compared to the effect of attractant management. In management scenarios with limited resources, the reduction of anthropogenic food sources should be prioritized above other strategies. In most cases, of course, multiple management strategies would be employed in tandem. No applied strategy will ever be 100% effective, but multiple strategies implemented in the same system can shore up weaknesses in each other. This was most evident in comparing effectiveness amongst model strategies; combining perimeter-configured attractant management, on its own an ineffective configuration, with randomly-configured increased deterrence being one of the most effective possible combinations. This case study provides an example of an agent-based modelling approach to evaluating diverse management strategies in a cost-effective, fast, and ethically sound manner. The variety and number of approaches explored in a simulated environment would not be possible in an experiential learning context. For example, the real-world time required to test even a small number of the replicates modelled above makes experimental testing unfeasible for comparing the effectiveness of one strategy over another. Additionally, the model's replication of the natural and urban configurations allows for tailored management regimes. The ability to simulate various strategy types and configurations multiple times in an identical landscape, which would prove challenging with any long-lived animal, can generate management plans customized to specific communities.

Although this model was configured for the Whistler landscape, the agent-based modelling approach used here could easily be adapted for other communities using appropriate landscape data and either the current bear movement parameters or movement parameters derived from bears local to the community of interest. Once constructed, such a model can be modified to simulate a variety of questions and scenarios specific to that community. Given appropriate GPS collar data, the model could also be modified to simulate movement and behaviour of other species as needed. This approach, using agent-based modelling to

test management strategies, can thereby be generally applicable to any community seeking to explore solutions for reducing local human-wildlife interactions.

Code availability

The model code and associated data analysis files are available for download from: <http://complexity.ok.ubc.ca/projects/whistler-bear-agent-based-model/>

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecolmodel.2021.109505](https://doi.org/10.1016/j.ecolmodel.2021.109505).

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